Fiscal Instruments and Substitutability in Markets for Indivisible Labor

October 2026

Abstract

How do taxes and subsidies contingent on a firm's hiring bundle or wages affect substitutability of demand for indivisible workers? In a finite transferable-utility labor market, we study fiscal changes to the employer's set valuation. A modular bundle transfer preserves an exchange condition sufficient for gross-substitutes demand. With two workers, a single inequality gives the exact threshold: fiscal complementarity cannot exceed the original substitution margin. A uniform singleton rebate, based only on a bound on production complementarity, restores substitutes under an explicit worst-case spending cap exactly when that cap covers the bound. We establish competitive equilibrium for all adjusted valuations satisfying the exchange condition by mapping the model to two precisely identified discrete-convexity theorems; the configuration LP consequently has an integral optimum. Separable wage-dependent transfers retain a demand-only result. A conditional bonus and a three-worker coverage valuation delimit the claims. Fiscal expenditure remains outside the welfare objective.

Keywords: indivisible labor; gross substitutes; fiscal policy; equilibrium.

1 Introduction

A firm choosing among named, indivisible workers can respond to one worker's higher wage by releasing another worker whose wage has not changed. Fiscal rules can themselves create or remove this interaction: an employer may receive a subsidy for a particular hiring bundle, or pay a tax determined by the wage it offers. We ask which features of these rules preserve or restore a demand condition, and which equilibrium conclusions can be established from it.

We represent a bundle-contingent net transfer as an addition to each firm's set valuation. The relevant property is the finite-set exchange condition in Definition 1. It implies gross-substitutes demand (Proposition 1). A modular transfer leaves every exchange comparison unchanged (Proposition 2). With two workers, the comparison reduces to a threshold on the transfer's cross difference (Proposition 3). Nonmodular rules can also satisfy the condition: a concave headcount payment does so for additive production (Proposition 9). Conversely, a bundle bonus can strictly violate it.

This manuscript is a post-unblind development revision of a historical reconstruction exercise. The general equilibrium step applies established mathematical theorems. The capped-rebate statement is a candidate result within this model and policy class, not a claim of priority or journal suitability.

For any bundle schedules producing EX valuations, Proposition 7 now establishes a clearing equilibrium. Its proof applies the real-valued convolution and intersection theorems of Murota [2022, Theorems 6.1(9) and 9.4], with all domain and price assumptions checked in Appendix F. The additive construction in Proposition 6 is an explicit special case. Proposition 8 adds a policy with limited valuation information and an ex-ante spending cap. Wage-dependent transfers have their own sufficient condition for gross-substitutes demand (Proposition 5); employer-specific effective wages still require a separate equilibrium argument.

2 Related literature

Kelso and Crawford [1982] develops a labor-market model with indivisible workers and a gross-substitutes requirement. Hatfield and Milgrom [2005] considers matching with contracts. Murota [2022] gives the exchange definition in Section 3.1, the binary/integer-vector identification in Section 4.1, and the convolution and intersection results used below. Thus the general EX equilibrium theorem here is an application of established discrete convex analysis, not a new existence theorem. Chen et al. [2023] studies competitive equilibrium with substitutes in more general discrete settings. Our policy calculations isolate the effect of specified transfers; we do not assert novelty against the full fiscal-policy literature, which has not been comprehensively surveyed here.

3 Environment and definitions

Let $F \neq \varnothing$ be a finite set of employers and N a finite set of named workers. A worker can be hired by at most one employer. An unhired worker has reservation utility zero. Employer f obtains gross, transferable-utility value $v_f(S) \in \mathbb{R}$ from hiring $S \subseteq N$, with $v_f(\varnothing) = 0$. A common vector of worker-specific wages is $w \in \mathbb{R}^N_+$. The employer pays $w(S) := \sum_{j \in S} w_j$, and every hired worker receives her wage. A net government payment $g_f(S, w)$ is positive for a subsidy and negative for a tax. The employer chooses S to maximize

$$U_f(S; w) = v_f(S) + g_f(S, w) - w(S). \tag{3.1}$$

Worker-specific reservation values are outside the environment. Proposition 8 adds an explicit information restriction and a worst-case transfer cap for one policy class; the general model still has no government financing or welfare primitive.

For a bundle schedule $g_f(S, w) = g_f(S)$, put $h_f(S) = v_f(S) + g_f(S)$. For any real price vector $p \in \mathbb{R}^N$, define

$$D_h(p) = \arg\max_{S \subseteq N} \{h(S) - p(S)\}. \tag{3.2}$$

Finiteness makes this correspondence nonempty. We use all real p when defining a property of h, although market wages are nonnegative.

Definition 1 (Exchange condition)**.** A set valuation h satisfies *EX* if, for every $X, Y \subseteq N$ and $i \in X \setminus Y$, either

$$h(X) + h(Y) \leq h(X \setminus \{i\}) + h(Y \cup \{i\}), \tag{3.3}$$

or there is $j \in Y \setminus X$ such that

$$\begin{aligned} h(X) + h(Y) \leq\ & h((X \setminus \{i\}) \cup \{j\}) \\ & + h((Y \setminus \{j\}) \cup \{i\}). \end{aligned} \tag{3.4}$$

This is the finite-set $M^\natural$ exchange form. No monotonicity or nonnegative-value assumption is imposed. The disjunction matters: a profitable exchange may replace the selected worker with one from the other bundle.

Definition 2 (Demand gross substitutes)**.** Demand for h satisfies *GS* if, whenever p' differs from $p \in \mathbb{R}^N$ only by $p'_k \geq p_k$, then for every $X \in D_h(p)$ there is $Y \in D_h(p')$ with $X \setminus \{k\} \subseteq Y$.

The definition permits ties and asks for *some* new maximizing bundle that retains every previously chosen worker whose price did not rise.

Definition 3 (Competitive equilibrium)**.** For bundle schedules, a competitive equilibrium is a wage vector $w \in \mathbb{R}^N_+$ and pairwise disjoint bundles $(S_f)_{f \in F}$ such that $S_f \in D_{h_f}(w)$ for every f, and $w_j = 0$ for every unhired worker j. A hired worker accepts the nonnegative wage; an unhired worker receives zero.

4 Results

Proposition 1 (Exchange implies demand substitutes)**.** *If h satisfies EX, then D_h satisfies GS.*

Appendix A proves this directly, including multiple demand maximizers. It is a sufficient bridge from valuation comparisons to choices; we do not invoke the converse for arbitrary N.

Proposition 2 (Modular fiscal schedule)**.** *Suppose $g_f(S) = c_f + \sum_{j \in S} b_{fj}$ for real c_f and b_{fj}. Then v_f satisfies EX if and only if $h_f = v_f + g_f$ satisfies EX. Consequently, if v_f satisfies EX, the adjusted demand is GS.*

Appendix B proves the claim. Each move or swap preserves the combined count of every worker across the two compared bundles. The c_f affects the level of profit, while positive or negative b_{fj} acts like a worker-specific price shift. This is preservation on the EX domain; modularity is not necessary for a given starting valuation.

Proposition 3 (Exact two-worker threshold)**.** *If $N = \{1, 2\}$, $h = v + g$ satisfies EX if and only if*

$$\underbrace{g(\{1,2\}) + g(\varnothing) - g(\{1\}) - g(\{2\})}_{\textit{fiscal cross difference}} \leq \underbrace{v(\{1\}) + v(\{2\}) - v(\{1,2\}) - v(\varnothing)}_{\textit{initial substitution margin}}. \tag{4.1}$$

For the all-real-price definition in Definition 2, this inequality is also necessary and sufficient for GS.

Appendix C proves both directions. The threshold allows a positive joint-hiring subsidy when the original valuation has enough substitution margin. If the original valuation is additive, the joint fiscal cross difference must be nonpositive. If original production has complementarity, a sufficiently negative fiscal cross difference can restore EX. Equality is allowed and can generate tied demand.

Proposition 4 (Implementation by a bundle schedule)**.** *For any normalized v_f and any normalized target valuation q_f satisfying EX, the schedule $g_f(S) = q_f(S) - v_f(S)$ makes $h_f = q_f$ and hence gives GS demand. In particular, an additive target $q_f(S) = \sum_{j \in S} a_{fj}$ can be implemented.*

This is the identity $v_f + g_f = q_f$ followed by Proposition 1. It is a feasibility statement about an unrestricted bundle schedule, not a cost or welfare result. Implementing it may require knowledge of every $v_f(S)$ and transfers of either sign.

Proposition 5 (Separable wage-dependent schedule)**.** *Suppose*

$$g_f(S, w) = g_f^0(S) - \sum_{j \in S} t_{fj}(w_j), \qquad q_{fj}(x) = x + t_{fj}(x), \tag{4.2}$$

and $h_f^0 = v_f + g_f^0$ satisfies EX. If every q_{fj} is weakly increasing on $\mathbb{R}_+$, employer f's demand satisfies GS with respect to actual nonnegative wages.

Appendix D proves this. Employer-specific effective prices are allowed. The proposition concerns demand alone: even if every employer has this property, we do not derive a common-wage equilibrium from it.

Proposition 6 (Additive adjusted values)**.** *If every adjusted valuation is $h_f(S) = \sum_{j \in S} a_{fj}$, with $a_{fj} \in \mathbb{R}$, a competitive equilibrium exists. Put $A_j = \max_{f \in F} a_{fj}$ and $w_j = \max\{0, A_j\}$. Assign each worker with $A_j > 0$ to one employer attaining A_j; leave all workers with $A_j \leq 0$ unassigned. This is an equilibrium. Its allocation maximizes adjusted private value, with*

$$\max_{(S_f)\ \textit{pairwise disjoint}} \sum_{f \in F} h_f(S_f) = \sum_{j \in N} \max\{0, A_j\}. \tag{4.3}$$

Appendix E verifies every firm's demand and market clearance, then proves (4.3). Because h_f includes government transfers, the objective in (4.3) is adjusted private value; it is not social surplus with fiscal costs accounted for.

Proposition 7 (Equilibrium for EX-adjusted values)**.** *If every h_f satisfies EX, a competitive equilibrium exists. Moreover, the configuration LP in Appendix F has an integral optimal solution.*

Appendix F proves this by a checked application of Murota [2022, Theorems 6.1(9) and 9.4]. The external intersection theorem, rather than a new integrality argument, carries the general existence step.

Proposition 8 (Information-limited, capped singleton rebate)**.** *Let $N = \{1, 2\}$, $|F| \geq 2$, and suppose the government knows only a common bound $\kappa \geq 0$ such that $v_f(\{1,2\}) + v_f(\varnothing) - v_f(\{1\}) - v_f(\{2\}) \leq \kappa$ for every f. Consider the uniform, nonnegative schedule $g_f(S) = r\mathbf{1}\{|S| = 1\}$, with $r \geq 0$, and require total transfers to be at most $B \geq 0$ for* every *feasible allocation. Within this policy class, a rebate that guarantees EX for every valuation profile consistent with the bound exists if and only if $B \geq \kappa$. When $B \geq \kappa$, $r = \kappa/2$ guarantees GS demand and a competitive equilibrium, with spending at most κ in every allocation.*

The necessity concerns uniform singleton rebates with an ex-ante cap, not all conceivable tax or subsidy schedules. Appendix G gives the full proof and a strict counterexample when $B < \kappa$.

5 Boundaries and counterexamples

Example 1 (A joint bonus destroys substitutes)**.** Let $N = \{1, 2\}$, $v(S) = |S|$, and set $g(\{1, 2\}) = 2$ and $g(S) = 0$ otherwise. Then $(h(\varnothing), h(\{1\}), h(\{2\}), h(\{1, 2\})) = (0, 1, 1, 4)$. The fiscal cross difference is 2, against an initial substitution margin of 0. At $p = (1.5, 1.5)$, profits from $(\varnothing, \{1\}, \{2\}, \{1, 2\})$ are $(0, -0.5, -0.5, 1)$, so demand is uniquely $\{1, 2\}$. Raising only p_1 to 4 gives profits $(0, -3, -0.5, -1.5)$ and uniquely empty demand. Worker 2 is dropped despite an unchanged price. This establishes failure of GS, not nonexistence of equilibrium.

Example 2 (Submodularity does not suffice)**.** Let $N = \{1, 2, 3\}$ and

$$h(S) = \mathbf{1}\{S \cap \{1, 3\} \neq \varnothing\} + \mathbf{1}\{S \cap \{2, 3\} \neq \varnothing\}.$$

Each summand has diminishing marginal returns, so h is submodular. The values are $h(\varnothing) = 0$, $h(\{1\}) = h(\{2\}) = 1$, and $h(S) = 2$ for every other nonempty S. For $X = \{1, 2\}$, $Y = \{3\}$, and $i = 1$, the original two-bundle value is 4, while both the move and the only swap yield 3: EX fails. At $p = (0.4, 0.4, 0.9)$, demand is uniquely $\{1, 2\}$; after only p_1 rises to 0.7, demand is uniquely $\{3\}$. The corresponding maximal profits are 1.2 and 1.1. Thus ordinary submodularity does not supply the multiperson exchange requirement or GS.

Proposition 9 (Nonmodular headcount transfers)**.** *Let $h(S) = \sum_{j \in S} a_j + \phi(|S|)$, where $\phi(0) = 0$, and write $d_k = \phi(k) - \phi(k-1)$. Then h satisfies EX if and only if $d_1 \geq d_2 \geq \cdots \geq d_{|N|}$.*

Appendix H proves the equivalence. Applied to an additive v, the proposition shows that a headcount transfer with diminishing increments can preserve substitution even when it is not modular. If an increment rises at any interior point, EX fails for this valuation class.

With a single worker, the EX and GS restrictions concerning retention of other workers are vacuous. For two workers, the threshold may bind and demand need not be unique. Finiteness guarantees maximizers and underlies the equilibrium construction; no infinite-worker conclusion follows. The necessity direction in Proposition 3 uses real prices and need not hold under a definition restricted only to nonnegative wages.

6 Discussion and limitations

The results distinguish three steps. A fiscal schedule changes an employer's set valuation or its effective wage vector. EX is a sufficient valuation condition for GS demand. For bundle schedules, the discrete-convexity argument establishes common-wage equilibrium for all EX-adjusted values. Demand GS under employer-specific wage wedges alone does not establish their common-wage equilibrium.

Unrestricted bundle correction in Proposition 4 is an extreme benchmark: it presumes access to the full array of employer values and permits transfers of arbitrary size. Proposition 8 instead uses one certified complementarity bound and an explicit worst-case spending cap, but its exact threshold applies only to two workers and uniform singleton rebates. The cap does not specify financing, incidence, or a social-welfare criterion. Neither the adjusted-value maximum nor a restored demand property ranks policies by social welfare. The model also abstracts from worker heterogeneity in reservation utility, wage bargaining, endogenous labor supply, and incentive or commitment constraints. Claims about those features require additional primitives and proofs.

7 Conclusion

Fiscal interaction across hiring bundles determines whether a policy uses up or restores a valuation's substitution margin. A modular schedule preserves every exchange comparison; the exact two-worker condition measures the permissible cross interaction. More complex markets require the full exchange condition. The general equilibrium result follows from established convolution and intersection theorems, while the capped singleton-rebate result states precisely what limited information and a worst-case spending ceiling permit in the two-worker case.

A Proof of Proposition 1

Fix p, p' that differ only in coordinate k, with $p'_k \geq p_k$, and choose $X \in D_h(p)$. If $k \notin X$, $h(X) - p'(X) = h(X) - p(X)$, whereas any bundle containing k weakly loses payoff and every other bundle keeps its payoff. Thus $X \in D_h(p')$, and $Y = X$ works.

Suppose $k \in X$. Among the nonempty finite set $D_h(p')$, choose Y maximizing $|Y \cap (X \setminus \{k\})|$. Assume, toward a contradiction, that $i \in (X \setminus \{k\}) \setminus Y$. Apply EX to (X, Y, i). In the move case let $X' = X \setminus \{i\}$ and $Y' = Y \cup \{i\}$. Moving i preserves its combined incidence in the two bundles; it also leaves k's incidence separately unchanged in X versus X' and Y versus Y'. All other prices are identical across p, p'. Subtracting prices from (3.3) therefore gives

$$[h(X) - p(X)] + [h(Y) - p'(Y)] \leq [h(X') - p(X')] + [h(Y') - p'(Y')]. \tag{A.1}$$

Because X maximizes at p, its bracket is at least the X' bracket. Hence the Y' bracket is at least the Y bracket. Since Y maximizes at p', Y' also maximizes. But Y' contains one more element of $X \setminus \{k\}$, contradicting the choice of Y.

In the swap case take $j \in Y \setminus X$ given by (3.4), and let $X' = (X \setminus \{i\}) \cup \{j\}$ and $Y' = (Y \setminus \{j\}) \cup \{i\}$. Here $i \neq k$ and $j \neq k$ because $k \in X$. The same combined-price cancellation gives (A.1). Optimality again makes Y' a new maximizer, while adding $i \in X \setminus \{k\}$ and removing $j \notin X$ increases the chosen intersection. This is a contradiction. Hence $X \setminus \{k\} \subseteq Y$, proving GS.

B Proof of Proposition 2

Write $m(S) = c + \sum_{j \in S} b_j$. In the move (3.3), the sum of the two modular terms is the same on each side: each side contains two copies of c, and for every worker the combined incidence across the two bundles is unchanged. The same calculation applies to a swap in (3.4). Thus, for each triple (X, Y, i) and each candidate move or swap, its exchange inequality holds for v precisely when it holds for $v + m$. Taking the disjunction over candidates and then all triples proves EX equivalence. Proposition 1 gives GS when EX holds.

C Proof of Proposition 3

With $N = \{1, 2\}$, the only nontrivial EX comparison is $X = \{1, 2\}$ and $Y = \varnothing$: the move requires

$$h(\{1, 2\}) + h(\varnothing) \leq h(\{1\}) + h(\{2\}). \tag{C.1}$$

There is no worker in $Y \setminus X$ to exchange. For the other triples, either moving i leaves the two-bundle sum unchanged or the two sets are disjoint singletons and swapping them leaves the sum unchanged. Hence (C.1) is necessary and sufficient for EX. Substitution of $h = v + g$ and rearrangement gives (4.1); Proposition 1 gives GS sufficiency.

To show necessity for GS over all real prices, suppose (C.1) fails strictly. Then

$$h(\{2\}) - h(\varnothing) < h(\{1,2\}) - h(\{1\}).$$

Choose p_2 strictly between these values. Choose p_1 sufficiently negative that $\{1,2\}$ strictly beats $\{2\}$ and $\varnothing$. It strictly beats $\{1\}$ because $p_2 < h(\{1,2\}) - h(\{1\})$. Thus it is the unique initial demand. Raise p_1 sufficiently high that both bundles containing worker 1 are strictly inferior to $\varnothing$. Since $p_2 > h(\{2\}) - h(\varnothing)$, singleton $\{2\}$ is also inferior to $\varnothing$. The new unique demand is empty, dropping worker 2 although only worker 1's price rose. GS fails.

D Proof of Proposition 5

For any $w \in \mathbb{R}^N_+$, substitute (4.2) into (3.1) to obtain

$$U_f(S; w) = h^0_f(S) - \sum_{j \in S} q_{fj}(w_j).$$

Consequently, $D_f(w) = D_{h^0_f}(q_f(w))$. If only w_k increases, every effective price other than q_{fk} is unchanged, while q_{fk} weakly increases. Proposition 1 applied to h^0_f supplies, for each old demand bundle, a new maximizing bundle retaining all members except possibly k. This is GS with respect to actual wages on the nonnegative domain.

E Proof of Proposition 6

Since F is finite and nonempty, each A_j is attained. The proposed wages are nonnegative and assignments are pairwise disjoint. If $A_j > 0$, $w_j = A_j \geq a_{fj}$ for every employer; if $A_j \leq 0$, $w_j = 0 \geq a_{fj}$. Thus $a_{fj} - w_j \leq 0$ for every (f, j), so every bundle yields employer f profit at most zero. The empty bundle yields zero. Every worker actually assigned to f satisfies $a_{fj} = A_j = w_j$, making S_f yield exactly zero. Hence $S_f \in D_{h_f}(w)$. Every unassigned worker has $A_j \leq 0$ and wage zero, giving market clearance.

For (4.3), an arbitrary feasible allocation assigns each j to at most one firm and obtains contribution at most A_j if assigned, or zero if not. Its total adjusted value is therefore at most $\sum_j \max\{0, A_j\}$. The constructed allocation assigns every j with $A_j > 0$ to an employer attaining A_j and leaves the remaining workers unassigned, reaching that bound. Negative and zero A_j cause no exception.

F Proof of Proposition 7 and configuration-LP integrality

If $N = \varnothing$, every firm chooses its sole bundle $\varnothing$, and the LP has its evident integral solution. Assume henceforth $|N| \geq 1$. Identify each $S \subseteq N$ with its binary incidence vector. Extend h_f to $\widehat{h}_f : \mathbb{Z}^N \to \mathbb{R} \cup \{-\infty\}$ by assigning $-\infty$ outside $\{0,1\}^N$. Definition 1 is exactly the set-function exchange axiom (3.1)–(3.3) of Murota [2022, Section 3.1]; the binary extension is

the integer-vector $M^\natural$-concave function of Section 4.1, equations (4.2)–(4.4), in that source. No monotonicity or integer-valued-payoff assumption enters this identification.

Introduce one dummy employer d with $h_d(S) = 0$ for every $S \subseteq N$. It satisfies EX because every admissible move has equality. Let $I = F \cup \{d\}$ and define the integer supremal convolution

$$H(z) = \max\Big\{\sum_{i\in I} \widehat{h}_i(z_i) : z_i \in \mathbb{Z}^N,\ \sum_{i\in I} z_i = z\Big\}.$$

Take the maximum of an empty feasible set to be $-\infty$. Every effective domain is the finite nonempty cube $\{0,1\}^N$. Thus the maximum, whenever feasible, is attained, and H is never $+\infty$. Repeated application of Murota [2022, Theorem 6.1(9)], which covers real-valued integer-vector $M^\natural$-concave functions, makes H $M^\natural$-concave. Put $e = (1,\dots,1)$ and let $J(z) = 0$ for $z = e$, $-\infty$ otherwise. The singleton-domain function J is also $M^\natural$-concave: its exchange axiom has no pair of distinct finite-domain points. The dummy employer alone can take all workers, so $H(e)$ is finite and $e \in \operatorname{dom} H \cap \operatorname{dom} J$. Since J is finite only at e, that point maximizes $H + J$. The real-valued $M^\natural$-concave intersection theorem, Murota [2022, Theorem 9.4], therefore supplies $p \in \mathbb{R}^N$ for which

$$H(e) - p \cdot e \geq H(z) - p \cdot z \qquad (z \in \mathbb{Z}^N). \tag{F.1}$$

Theorem 9.4 supplies the supporting price, while Theorem 6.1(9) supplies the required closure; their domain and real-valued hypotheses have been checked above.

Choose a maximizing decomposition $(S_i)_{i\in I}$ of $H(e)$. Because $\sum_i \mathbf{1}_{S_i} = e$, the bundles partition N. Taking the maximum over all z and then all decompositions shows

$$\max_z\{H(z) - p \cdot z\} = \sum_{i\in I} \max_{T\subseteq N}\{h_i(T) - p(T)\}.$$

By (F.1), the chosen decomposition attains the left side. Its sum of individual payoffs therefore attains the sum of individual maxima; each S_i must be in $D_{h_i}(p)$. In particular, the dummy's zero valuation implies $p_j \geq 0$ when $j \notin S_d$ (adding j would otherwise improve its payoff) and $p_j \leq 0$ when $j \in S_d$ (removing j would otherwise improve it). Set $w_j = p_j$ for $j \notin S_d$ and $w_j = 0$ for $j \in S_d$. Then $w \geq 0$, and the workers held by the dummy are precisely those unassigned to real employers and have zero wage. For a real employer f, w only raises prices on S_d, which is disjoint from S_f. Thus $h_f(S_f) - w(S_f) = h_f(S_f) - p(S_f)$, while every alternative T has $h_f(T) - w(T) \leq h_f(T) - p(T)$. Hence $S_f \in D_{h_f}(w)$, proving equilibrium.

For completeness, this equilibrium also closes the configuration-LP obligation. Define nonnegative configuration variables x_{fS} for every real employer f and $S \subseteq N$. The primal is

$$\begin{aligned}
\max_{x\geq 0} \quad & \sum_{f\in F}\sum_{S\subseteq N} h_f(S)x_{fS} && && \text{(F.2)}\\
\text{subject to} \quad & \sum_{S\subseteq N} x_{fS} = 1 && (f \in F),\\
& \sum_{f\in F}\sum_{S\ni j} x_{fS} \leq 1 && (j \in N).
\end{aligned}$$

The dual, with unrestricted employer variables u_f and nonnegative worker variables w_j, is

$$\begin{aligned} \min_{u\in\mathbb{R}^F, w\geq 0} \quad & \sum_{f\in F} u_f + \sum_{j\in N} w_j \\ \text{subject to} \quad & u_f + w(S) \geq h_f(S) \qquad\qquad (f\in F,\ S\subseteq N). \end{aligned} \tag{F.3}$$

Set $x_{fS_f} = 1$ and all other configuration variables to zero. This is primal feasible and integral. Set $u_f = \max_T\{h_f(T) - w(T)\}$. Then (u, w) is dual feasible; demand optimality gives $h_f(S_f) = u_f + w(S_f)$. Since wages of unassigned workers vanish,

$$\sum_f h_f(S_f) = \sum_f u_f + \sum_{j\in N} w_j.$$

The feasible primal and dual objectives coincide, so both are optimal by weak duality. In particular, the LP has an integral optimum. This argument also shows exactly how the external intersection theorem fills the former integrality gap.

G Proof and counterexample for Proposition 8

For any feasible allocation, at most two employers can each hire one worker, so the singleton rebate pays at most $2r$. Since $|F| \geq 2$, an allocation that gives worker 1 to one employer and worker 2 to another attains $2r$; hence the ex-ante cap is equivalent to $2r \leq B$. For each employer the fiscal cross difference in (4.1) is $-2r$, while the production substitution margin is at least $-\kappa$ by hypothesis. Thus $2r \geq \kappa$ guarantees EX for every admissible valuation profile, including the equality case. Proposition 1 gives GS and Proposition 7 gives equilibrium. Taking $r = \kappa/2$ obeys the cap exactly when $B \geq \kappa$.

Conversely, suppose $B < \kappa$. Every admissible rebate has $2r \leq B < \kappa$. Give one employer the normalized valuation $v(\{1\}) = v(\{2\}) = 0$ and $v(\{1,2\}) = \kappa$; give every other employer the zero valuation. The information bound holds, but the first employer has $h(\{1\}) = h(\{2\}) = r$, $h(\{1,2\}) = \kappa$, and $h(\varnothing) = 0$. Hence $h(\{1,2\}) + h(\varnothing) > h(\{1\}) + h(\{2\})$, so EX fails. Proposition 3's real-price necessity argument also gives a strict GS failure: choose p_2 with $r < p_2 < \kappa - r$, choose p_1 sufficiently negative so $\{1,2\}$ is uniquely demanded, then raise only p_1 sufficiently high so the empty set is uniquely demanded. Worker 2 is lost at an unchanged price. This proves necessity of $B \geq \kappa$ for a uniform singleton rebate robust over the stated information class; it does not assert equilibrium nonexistence in the counterexample.

H Proof of Proposition 9

Write $m(S) = \sum_{j\in S} a_j$ and $n = |N|$. Suppose $d_1 \geq \cdots \geq d_n$, and fix X, Y and $i \in X \setminus Y$. If $|X| > |Y|$, move i from X to Y. The sum of modular terms is unchanged, and the change in headcount terms, new minus old, equals

$$\phi(|X|-1) + \phi(|Y|+1) - \phi(|X|) - \phi(|Y|) = d_{|Y|+1} - d_{|X|} \geq 0,$$

because $|Y| + 1 \leq |X|$. Thus (3.3) holds. If $|X| \leq |Y|$, the facts that $i \in X \setminus Y$ and $|X| \leq |Y|$ imply $Y \setminus X \neq \varnothing$. Choose any j there and swap i, j. Both bundle sizes and the combined

incidence of each worker stay fixed, so both the headcount and modular sums are unchanged; (3.4) holds. Hence h satisfies EX.

Conversely, if $d_{k+1} > d_k$ for some $1 \leq k < n$, choose X of size $k+1$, a subset $Y \subset X$ of size $k-1$, and $i \in X \setminus Y$. There can be no swap because $Y \setminus X = \varnothing$. The new-minus-old value in the move is

$$2\phi(k) - \phi(k+1) - \phi(k-1) = d_k - d_{k+1} < 0;$$

the modular sum cancels. The required move inequality fails, so EX fails. This proves necessity.

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